DongYu Weng

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Education

Carleton College — Northfield, MN
Expected Bachelor of Arts in Physics and Computer Science — *Expected June 2028*

Research

Integrated Optofluidic Devices — Sep. 2026 – Present
Carleton College (Prof. Martha-Elizabeth Baylor) — *Northfield, MN*

- Characterize photopolymer resins used to fabricate integrated optofluidic devices that combine optical waveguides and fluidic channels on a single chip
- Build apparatus to measure resin properties across formulations, including refractive index, optical loss per unit length, and stiffness
- Working toward a reference table of materials so engineers can choose the right resin for applications ranging from wearables to harsh environments

Optics Education Research — Jan. 2026 – June 2026
Carleton College (Prof. Martha-Elizabeth Baylor) — *Northfield, MN*

- Designed and implemented accessible, low-cost optics activities to engage underrepresented communities in foundational physics concepts
- Assessed student comprehension through activity-based evaluations and refined materials to improve engagement and conceptual understanding

Experience

Makerspace Manager — Sep. 2024 – Present
Carleton College — *Northfield, MN*

- Assist students and faculty members in projects ranging from sticker making to robot building

Physics Lab Assistant — Sep. 2026 – Present
Carleton College — *Northfield, MN*

- Provide hands-on instruction in physics lab sections, guiding students through experimental setup, data collection, and analysis
- Explain the physics behind each experiment and troubleshoot equipment so students can work through problems independently

Engineering Design Associate — May 2025 – Aug. 2026
Fuzhou Changhui Light Steel Structure Building Company — *Fuzhou, China*

- Converted 2D building schematics into 3D models using CAD software, improving visualization and structural accuracy
- Collaborated with engineers and designers to ensure compliance with project specifications and safety standards

Data Analysis Intern — Mar. 2024 – June 2024
Office of The State Superintendent of Education — *Washington, DC*

- Researched the effects of chronic absenteeism and truancy within DC schools culminating in a presentation to the public

Barista — July 2023 – May 2024
Luna Hall DC — *Washington, DC*

- Served as cashier and prepared specialty boba beverages in a fast-paced customer service environment

Certifications and Technical Skills

Google: Cyber Security, Data Advanced Analysis Specialization, IT Automation with Python Specialization
Georgetown School of Continuing Studies: Cybersecurity: Protecting Our Digital Future
Languages: Python, C/C++, Kotlin, HTML/CSS, Scheme
Developer Tools: Git, Docker, VS Code, Visual Studio, Eclipse, Autodesk Fusion 360, Microsoft Excel